

$$f(x) = \frac{x+1}{x} = 1 + \frac{1}{x}$$

$$D_f = \mathbb{R} \setminus \{0\}$$

$$\text{Zeros: } f(x)=0 \Leftrightarrow x+1=0 \Leftrightarrow x=-1$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{x+1}{x} = \frac{\infty}{\infty} = \lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right) = 1$$

$$f'(x) = \left(\frac{x+1}{x}\right)' = \frac{(x+1)'x - x'(x+1)}{x^2} = \frac{x - x - 1}{x^2} = \frac{-1}{x^2}$$

$$\text{Zeros: } f'(x) = 0 \Leftrightarrow \text{Não tem zeros}$$

$$f''(x) = (-1/x^2)' = (-x^{-2})' = (-2)x^{-2-1} = -2x^{-3} = -2/x^3 \quad \text{ou} \quad (-1/x^2)' = \frac{(-1)'x^2 - (x^2)'(-1)}{x^4} = \frac{0+2x}{x^4} = \frac{2}{x^3}$$

$$\text{Zeros: } f''(x) = 0 \Leftrightarrow \text{Não tem zeros}$$

x	$-\infty$		-1		0		$-\infty$
f(x)	1	+	0	-	~	+	1
f'(x)		- ↘		- ↘	~	- ↘	
f''(x)		- ∪		- ∪	~	+ ∩	
		↘	Z	↘	~	↘	

